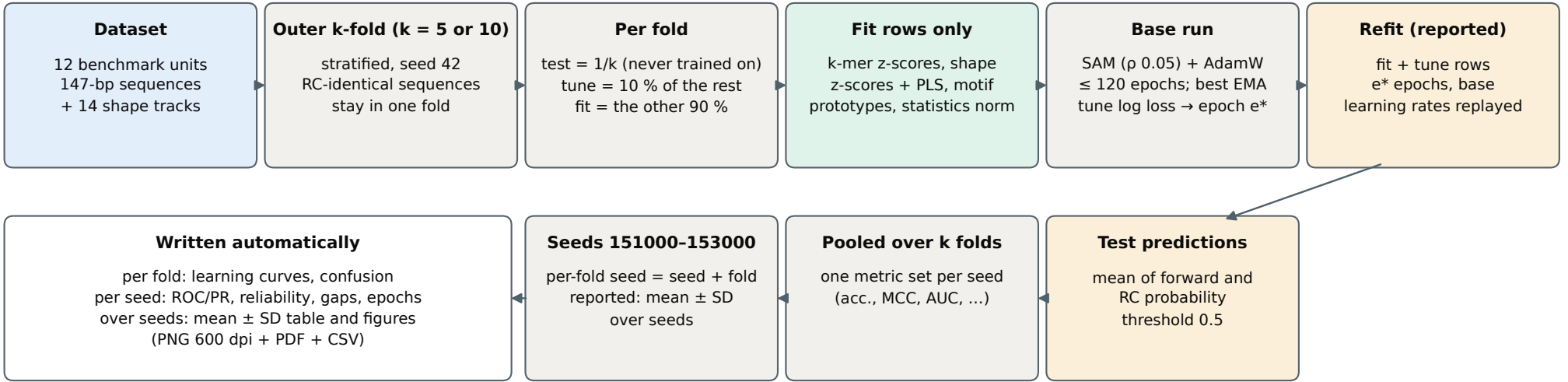
Dataset
12 benchmark units
147-bp sequences
+ 14 shape tracks
Outer k-fold (k = 5 or 10)
stratified, seed 42
RC-identical sequences
stay in one fold
Per fold
test = 1/k (never trained on)
tune = 10 % of the rest
fit = the other 90 %
Fit rows only
k-mer z-scores, shape
z-scores + PLS, motif
prototypes, statistics norm
Base run
SAM (ρ 0.05) + AdamW
≤ 120 epochs; best EMA
tune log loss → epoch e*
Refit (reported)
fit + tune rows
e* epochs, base
learning rates replayed
Test predictions
mean of forward and
RC probability
threshold 0.5
Pooled over k folds
one metric set per seed
(acc., MCC, AUC, …)
Seeds 151000–153000
per-fold seed = seed + fold
reported: mean ± SD
over seeds
Written automatically
per fold: learning curves, confusion
per seed: ROC/PR, reliability, gaps, epochs
over seeds: mean ± SD table and figures
(PNG 600 dpi + PDF + CSV)